Incentive Compatibility and Payoff Robustness in Multiplayer Quantum Games: Theory and Hardware Evidence

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***Abstract*—We study how entangler family, player count, and implementation noise affect payoff and incentives in an N-player Eisert–Wilkens–Lewenstein allocation game. Five families define an exact simulation landscape, with graph-role asymmetry distinguished from circuit-order and gate-budget effects. For the global GHZ operator, we derive a family of cooperative phase strategies and an exact restricted-menu incentive boundary. The original π/N branch fails the Hawk-deviation test for $N \geq 4$, whereas a separately defined branch is a strict restricted-menu equilibrium for every $N \geq 2$ at maximal entanglement. Both admit an explicit profitable unrestricted SU(2) deviation. A four-product-state representation permits exact classical payoff evaluation through $N = 128$ without dense state enumeration. Existing IBM Heron r2 experiments evaluate the original branch through seven players, including registered Hawk-deviation tests at $N = 3$ and failed out-of-sample noise-model predictions. Mean-payoff retention is intrinsically forgiving: uniform independent bits retain 99.22% of the ideal analytic-baseline gap at $N = 7$. A retrospective sensitivity analysis that charges for partial conflicts gives raw $N = 7$ retention of 93.23%, against a 75% uniform comparator. These results separate scalable welfare evaluation from incentive compatibility and physical implementation. Two new all-player hardware batches support the registered incentive criterion at $N = 4, 6$ in both runs, but fail the overall five-size test; $N = 7$ remains unresolved. The trading interpretation is illustrative rather than an economic validation.**

***Index Terms*—quantum games, Nash equilibrium, EWL protocol, entanglement topology, NISQ, error mitigation, quantum finance**

I. INTRODUCTION

Financial markets are coordination problems with adversarial payoffs. When traders compete over a scarce asset, each chooses between an aggressive position (capturing the full value if unopposed) and a cooperative one that shares value reliably. The Hawk–Dove game formalizes this tension. Under the live convention used throughout this work, V=4 and C=3, Hawk strictly dominates Dove in the two-player payoff matrix: it pays more against either opposing action, so mutual Hawk is the unique classical pure Nash equilibrium. Yet mutual Dove pays each player $V/2 = 2$, whereas mutual Hawk pays only $(V - C)/2 = 0.5$. The resulting gap isolates the mechanism-design problem: individually rational aggression destroys value that cooperation would preserve. Classical remedies are external to the game, including trust between counterparties, a trusted intermediary, or regulation that taxes defection.

Quantum game theory offers a different lever. In the Eisert–Wilkens–Lewenstein (EWL) protocol [1], strategies live in SU(2) and players' qubits are entangled before any move is made; the correlation structure of the shared state then changes what is self-enforcing. Within its restricted strategy space, the two-player result is a quantum Nash equilibrium that is simultaneously symmetric and Pareto-optimal within the stipulated protocol. Preparation, joint decoding, and strategy-menu enforcement remain mechanism assumptions. Carbon-trading and order-book architectures provide motivating analogies; we do not establish replacement of regulatory or trust mechanisms.

Khan et al. [2] developed quantum trading models and reported ion-trap experiments with three to six players in a Prisoner's-Dilemma variant. Our extension concerns a different allocation rule and controlled comparisons of entangler families, incentives, and implementation effects. Within our simple pairwise graph convention, two players admit only one edge. Real markets are networks: the *topology* of who is correlated with whom is a design variable, and each topology induces a different interference structure, different equilibria, and a different response to decoherence. Varsamis et al. [3] studied N-player entanglement operators and scalability on IBM hardware; our contribution is orthogonal: we fix a financially interpreted game and chart how *topology* $\times$ *player count* $\times$ *noise* determine both the magnitude and the equilibrium status of the quantum advantage, then test the resulting noise model against registered predictions on hardware.

The main contributions of this work are as follows:

1) **Incentive boundary and scalable evaluation.** We derive the GHZ cooperative phase branches and their exact

restricted-menu deviation condition. A separate branch satisfies that condition for every $N \geq 2$ at maximal entanglement, while an explicit unrestricted SU(2) deviation defeats it. A four-product-state representation enables exact classical payoff evaluation through $N = 128$, distinct from physical player-count scaling.
2) **Advantage landscape.** The complete $(N, \text{topology})$ circuit-relative payoff-gap map for $N = 2$–6 over five topologies. For GHZ, the fixed cooperative profile $(Q_N, \ldots, Q_N)$ pays exactly V/N and passes the restricted $\{D, H, Q_N\}$ deviation check at $N = 2, 3$, not against arbitrary SU(2) gates; ring and fully connected also pass that restricted check at $N = 3$. Other topology cells may or may not attain V/N.
3) **Noise-robustness dichotomy.** Under depolarizing noise the restricted-menu equilibrium criterion and circuit-relative payoff-gap criterion order topologies differently; noise also breaks player symmetry, producing position-locked per-player disadvantage that follows the entangler's gate ordering (shown by a wiring-permutation experiment). Matched-gate-count and compilation controls reduce gate-budget confounding within the XX-product family (GHZ/ring/star/fully connected), whose normalized decay is similar in the control table, while absolute GHZ–W ordering can remain because their zero-noise payoff gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. In an exploratory W $N{=}4$ finite-round pilot, no tested setting produced a beneficial converged fairness repair.
4) **Predictive hardware validation.** An $N = 3, 4, 5$ scaling experiment on ibm_fez (one pinned chain, one batch job) with tensored readout mitigation and cz-fold zero-noise extrapolation retains 98–99% of the ideal advantage $3/N$; the one-parameter depolarizing fit misses the retrospective source-run mitigated-fold-1 endpoints at $N{=}4$ by 0.0043 (2.14σ) and $N{=}5$ by 0.0101 (5.17σ). In the prospective repeat the conditional $N{=}4$ endpoint passes and the $N{=}5$ deficit repeats, observationally rather than as a registered sign test. The CZ-exponential baseline scores lowest among the registered candidates but still misses $N{=}5$ outside the shared ranking yardstick. We identify the mechanism that makes the advantage more noise-robust than the state: first-order insensitivity of the mean payoff to single bit-flips.
5) **Reproducible pipeline.** A versioned research harness with circuit-identity gates, a mandatory device-model dress rehearsal before any hardware submission, and crash-safe job recovery.

The rest of the paper is organised as follows. Section II reviews related work, and Section III defines the evaluation questions and evidence boundaries. Section IV develops the mathematical framework, followed by computation and validation methods in Section V. Section VI reports simulation and hardware results. Section VII discusses their interpretation, scalability, and limitations. Section VIII states the technical differentiation, followed by data and code availability and the conclusion in Section IX.

II. Related Work

A. Quantum games and the EWL protocol

The EWL quantization [1] embeds player strategies in SU(2) and entangles the player register before any strategy is applied. Its equilibrium properties depend on both the entangler and the permitted strategy set. The original two-player restricted-strategy result admits a symmetric, Pareto-optimal equilibrium at maximal entanglement, $\gamma = \pi/2$; enlarging the strategy space can invalidate that equilibrium [1], [4]. The protocol is the base layer of this work: we keep its structure and vary what it entangles.

B. Multiplayer extensions and entanglement structure

Analytic multiplayer cooperation equilibria also occur in the quantum volunteer's dilemma [5]. Its payoff and classical comparator differ from the allocation game studied here; an extension to multiple players alone is therefore not our novelty claim. Benjamin and Hayden [6] give a multiplayer quantum-game construction, including minority-game and Prisoner's-Dilemma examples. We adopt its circuit framework; the shared-resource allocation payoff in Sec. IV is the separately specified model studied here. Chappell, Iqbal, and Abbott [7] compared GHZ and W entanglement in N-player quantum games in an EPR setting, but in simulation and without a financial payoff structure or a hardware validation; our contribution is to carry that GHZ-versus-W question through a concrete trading game to a noise-model-tested result on a real device.

C. Quantum advantage in trading games

Khan et al. [2] developed two-player trading models and reported multiplayer ion-trap experiments with three to six traders using a Prisoner's-Dilemma payoff rule. We distinguish our allocation game and entangler-family controls from that prior multiplayer hardware demonstration. Within the simple pairwise graph convention, two players admit only one edge: the question of which correlation graph a market designer should choose only appears at $N \geq 3$, and that is the question this paper addresses.

D. N-player hardware scalability and error mitigation

Varsamis et al. [3] study N-player entanglement operators and their scalability on IBM hardware generically, characterising the operators themselves across devices. We instead fix a single game (the financial Hawk–Dove) and map its *advantage landscape*: the topology-dependent equilibrium/payoff dichotomy, the noise-induced position-locked unfairness, and a registered predictive noise model whose out-of-sample predictions we test on hardware. On the mitigation side we adopt two standard NISQ techniques rather than proposing new ones: tensored confusion-matrix inversion for readout errors [8], [9] and zero-noise extrapolation by local gate folding [10], [11]; their use in this pipeline is specified in Sec. VI-B.

III. Evaluation Questions and Evidence Boundaries

The study addresses four questions with different units of evidence. First, which cooperative phase profiles resist unilateral deviations in the specified game? Second, how do entangler families change payoffs and fairness at a fixed strategy? Third, which apparent robustness differences survive controls for circuit implementation? Fourth, which ideal properties remain supportable on a finite noisy processor? Table I states the operational test and interpretation of each question.

A. Mathematical identities and empirical estimands

The phase boundary and four-product-state representation are algebraic properties of the ideal GHZ protocol. Finite floating-point tests check their implementation but do not establish the all-N statements by themselves. Hardware probabilities instead estimate repeated circuit outcomes. The mean payoff, each player's payoff, the worst unilateral gain, and the all-zero population are distinct estimands. In particular, high mean payoff cannot substitute for a player-level incentive test.

B. Evidence hierarchy and negative findings

Historical runs, retrospective reweighting, and prospectively registered tests retain their separate status. Failed noise predictions remain in the record. The alternative-phase repeat is judged under the same rule as the first batch, without pooling to reverse a failed acceptance test. No result on a finite device is promoted to a statement about unlimited physical scaling. Compact classical evaluation is reported as an analysis tool, while physical player count remains the number of simultaneously represented qubits in the executed protocol.

IV. Proposed N-Player Quantum Hawk–Dove Framework

A. Notation and the Classical N-Player Payoff Tensor

Before quantizing the game, we need a payoff rule that assigns a value to every one of the 2^N classical outcomes and preserves each player's identity.

We write $\mathcal{N} = \{1, \ldots, N\}$ for the set of players. Every player $j \in \mathcal{N}$ chooses one of two actions, and we record a joint choice as a bitstring $x = (x_1, \ldots, x_N) \in \{0,1\}^N$, where $x_j = 1$ means player j plays Hawk and $x_j = 0$ means player j plays Dove. Hawk is the aggressive action: it takes the whole contested value when unopposed, but destroys value when it meets itself. Dove is the cooperative action, which shares value reliably. Two constants set the stakes: the contested resource is worth $V > 0$, and a confrontation between Hawks costs $C > 0$. This is the two-strategy value-and-cost structure of the classical Hawk–Dove game [12], carried to N players inside the general framework of symmetric multi-player matrix games [13]. We fix $V{=}4,\ C{=}3$ for every numerical statement in this paper (the repository's live convention, `src/config.py`), and keep V and C symbolic wherever a result does not depend on their values.

One statistic drives every classical payoff: the Hawk count

$$k(x) = \sum_{j=1}^{N} x_j \in \{0, 1, \ldots, N\}, \tag{1}$$

which counts how many players choose the aggressive action at outcome x. The payoff tensor then assigns to each player j and each outcome x the value

$$P_j(x) = \begin{cases} V/N, & k(x) = 0, \\ V/k(x), & 0 < k(x) < N \text{ and } x_j = 1, \\ 0, & 0 < k(x) < N \text{ and } x_j = 0, \\ (V-C)/N, & k(x) = N. \end{cases} \tag{2}$$

In words, the four branches of the payoff tensor, Eq. (2), say that when nobody fights the players split the asset evenly; when some but not all fight, the Hawks split the asset among themselves and the Doves receive nothing; and when everybody fights, the players split what the conflict leaves behind. We write $P(x) = (P_1(x), \ldots, P_N(x))$ for the profile of per-outcome payoffs, and we reserve the term *payoff vector* for the vector of expected player payoffs introduced later in this section with the comparison and fairness metrics.

An $N = 4$ example makes the branches concrete. At the all-Dove outcome $x = 0000$ the Hawk count is $k = 0$, so every player receives $V/4$. At the one-Hawk outcome $x = 1000$ the Hawk count is $k = 1$: the lone Hawk receives $V/1 = V$ and each of the three Doves receives 0. At the all-Hawk outcome $x = 1111$ the Hawk count is $k = 4$, so every player receives $(V - C)/4$. Each branch keeps the player index, because P_j depends on the player's own action x_j as well as on the aggregate $k(x)$: two players at the same outcome need not be paid the same. That per-player resolution is what later lets us measure how a quantum protocol redistributes value between graph positions rather than only how much value it creates in total.

Two properties of the payoff tensor, Eq. (2), carry through the rest of the paper. It is defined for every N and every one of the 2^N outcomes, so scaling the player count needs no special cases; and it depends on an outcome only through the pair $(k(x), x_j)$, which is what makes the welfare geometry of the next subsection tractable. Extending a two-player conflict game to N players under a shared entangling operation follows the multiplayer quantum-game construction of Benjamin and Hayden, who give worked $N = 3$ and $N = 4$ cases [6]; Eq. (2) is the closed form for general N that this paper uses throughout. With the classical payoffs fixed, we can ask what they imply about equilibrium and total welfare.

TABLE I
EVALUATION QUESTIONS, EVIDENCE LAYERS, AND INTERPRETATION BOUNDARIES.

Question	Operational test	Evidence	Interpretation permitted
Which phases support cooperation?	Compare every named unilateral strategy with the cooperative payoff.	Phase-boundary proof, unrestricted witness, and dense-circuit cross-checks.	Exact ideal statements for the specified operator and menu; no unrestricted equilibrium claim.
How does the entangler affect payoffs?	Evaluate the same fixed strategy and enumerate finite-menu best replies.	Five-family noiseless landscape, per-player vectors, and graph-symmetry checks.	A fixed-strategy comparison; not each family's globally optimal play.
What determines noise robustness?	Compare payoff gaps and equilibrium status; vary compilation and match gate counts.	Gate-level sweeps, implementation controls, and wiring permutations.	Production-circuit robustness; geometry and synthesis are not interchangeable.
What survives on hardware?	Test all-player deviation gaps under a frozen confidence rule; inspect welfare and population separately.	Registered circuits, raw counts, calibration records, and repeated batches.	Bounded implemented-game evidence; no distributed-market deployment claim.

B. Classical Equilibrium and Welfare Geometry

The payoff tensor says what each outcome pays, but not which outcome self-interested players reach, nor what the group holds in total once they get there; a quantum protocol can only be said to improve on the classical game once both are settled.

Hawk strictly dominates Dove at the live parameters. Take player j facing m opponents who play Hawk, $0 \leq m \leq N-1$, and read off the branches of the payoff tensor, Eq. (2), that player j can choose between. Playing Hawk raises the Hawk count to $m+1$ and pays $V/(m+1)$ when $m < N-1$, or $(V-C)/N$ when $m = N-1$. Playing Dove leaves the Hawk count at m and pays V/N when $m = 0$, or 0 otherwise. The three comparisons are V against V/N at $m = 0$; $V/(m+1) > 0$ against 0 for $0 < m < N-1$; and $(V-C)/N$ against 0 at $m = N-1$. The first is strict for every $N \geq 2$, the second always, and the third exactly when $V > C$. At $V{=}4,\ C{=}3$ all three are strict, so Hawk pays more than Dove against every opponent configuration. The condition that produces dominance is $V > C$, not the Hawk–Dove structure itself, which is why we scope the statement to the live parameters.

Strict dominance fixes the classical benchmark. Since Hawk is the better reply to every opponent configuration, the all-Hawk outcome $x = 1^N$ is the unique pure-strategy Nash equilibrium of the classical game [14]: no player gains by unilaterally switching to Dove, and no other profile survives one round of best replies. At $V{=}4,\ C{=}3$ it pays $(V-C)/N = 1/N$ per player. That $1/N$ is the classical baseline against which this paper measures every advantage.

All-Dove pays $V/N = 4/N$ per player and Pareto-dominates that equilibrium, so the classical game is a social dilemma in the standard sense: the strictly dominant action, taken by everyone, leaves everyone worse off than mutual cooperation [15]. At $N = 2$ the payoff tensor specializes to the familiar two-player conflict matrix, paying $V/2$ to each player under mutual Dove, $(V, 0)$ under one-sided aggression, and $(V-C)/2$ to each under mutual Hawk. The interior mixed equilibrium usually associated with the Hawk–Dove game [12] is unavailable at these parameters: indifference between the two actions requires a Hawk probability of V/C, and $V/C = 4/3$ falls outside the probability simplex whenever $V > C$. The all-Hawk pure equilibrium is therefore the only classical baseline this paper needs.

Individual payoffs move sharply between outcomes, but total value does not.

Proposition 1 (Outcome-level welfare)**.** *For every $x \in \{0,1\}^N$,*

$$\sum_{j=1}^{N} P_j(x) = V - C\,\mathbf{1}_{\{x=1^N\}}. \tag{3}$$

Proof. Split on the Hawk count $k(x)$. If $k(x) = 0$, every player is a Dove paid V/N and the N terms sum to V. If $0 < k(x) < N$, the $k(x)$ Hawks are each paid $V/k(x)$ and the $N - k(x)$ Doves are each paid 0, so the sum is $k(x) \cdot V/k(x) = V$ again. If $k(x) = N$, every player is a Hawk paid $(V-C)/N$ and the sum is $V - C$. The indicator selects that last case, the only one in which value is destroyed rather than redistributed. □

The welfare identity, Eq. (3), controls mean payoff under any outcome distribution. Let p be the distribution over the 2^N outcomes produced by whatever procedure the players follow — classical randomization now, a quantum circuit from the next subsection onward — and let $\bar{\pi}$ be the resulting mean payoff per player, the average of the per-player expected payoffs defined with the metrics later in this section. Averaging the welfare identity over p gives

$$\bar{\pi} = \frac{1}{N}\sum_{x} p(x) \sum_{j=1}^{N} P_j(x) = \frac{V}{N} - \frac{C}{N}\,p(1^N). \tag{4}$$

In plain terms, mean payoff depends on the outcome distribution through one number: the probability of the all-Hawk outcome. Every other outcome, however unequal its per-player split, redistributes the same total value V. A protocol can therefore raise mean welfare only by suppressing $p(1^N)$, and it can do so while leaving individual players very unequally paid — a gap between mean and fairness that Sec. VI-A measures directly. What such a protocol looks like is the subject of the next subsection.

C. Quantum Strategies and the EWL Protocol

Classical play reaches all-Hawk and destroys C of the contested value, so the question is what changes when the players' actions are carried by qubits rather than by coins.

The Eisert–Wilkens–Lewenstein (EWL) protocol answers it in three steps [1]. One qubit is issued to each player and the register is entangled by a shared operation agreed before play. Each player then acts locally and privately on their own qubit, exactly as a classical player acts privately on their own choice. Finally the entangling operation is undone and the register is measured, which returns one of the 2^N classical outcomes of Sec. IV-A and pays the players by the payoff tensor, Eq. (2). The players never share a signal during play; all correlation is set up in advance and released by the measurement.

A player's action is a single-qubit unitary. We take the full special unitary group, writing a strategy in the three-parameter form

$$U(\theta,\alpha,\beta)=\begin{pmatrix} e^{i\alpha}\cos(\theta/2) & ie^{i\beta}\sin(\theta/2) \\ ie^{-i\beta}\sin(\theta/2) & e^{-i\alpha}\cos(\theta/2)\end{pmatrix}, \qquad (5)$$

with $\theta \in [0,\pi]$ and $\alpha,\beta \in [-\pi,\pi]$. The angle θ controls how much a player rotates between the two classical actions, while α and β set phases that have no classical counterpart and are the whole source of the protocol's extra structure.

Three named strategies recur throughout the paper:

$$\begin{gathered} D=U(0,0,0), \qquad H=U(\pi,0,0), \\ Q_N=U(0,\pi/N,\pi/N). \end{gathered} \qquad (6)$$

Here D is the identity and leaves a player's qubit in the Dove state, H equals $i\sigma_x$ and flips it to Hawk (the leading factor i is a global phase and is unobservable), and Q_N is the phase-only strategy $\operatorname{diag}(e^{i\pi/N}, e^{-i\pi/N})$, which has no classical counterpart at all. The π/N scaling makes Q_N GHZ-derived: it is the phase that the GHZ entangler's two branches require, so the familiar two-player $Q = U(0,\pi/2,\pi/2)$ is the $N = 2$ case of one family rather than a universal quantum strategy. We hold Q_N fixed when we change entangler family, so that a family comparison changes the correlation structure and nothing else.

Writing U_j for the strategy of player j and $J_T(\gamma)$ for the entangling operation of family T at entanglement angle γ, the protocol produces

$$|\psi_{\text{out}}\rangle = J_T^{\dagger}(\gamma)\left(\bigotimes_{j=1}^{N} U_j\right) J_T(\gamma)|0^N\rangle, \qquad (7)$$

and measuring in the computational basis gives each classical outcome the probability

$$p_T(x \mid \boldsymbol{U}) = |\langle x|\psi_{\text{out}}\rangle|^2, \qquad (8)$$

where x_j is the measured value of player j's qubit, matching the bitstring convention of Sec. IV-A. The two equations say that the protocol is a map from a strategy profile to a distribution over the same classical outcomes the classical game had; the payoff rule is untouched, and every result in this paper comes from moving probability between those outcomes. The entanglement angle γ controls the interpolation used to correlate the register before the players act: every family we study satisfies $J_T(0) = I$, the identity on all N qubits, so $\gamma = 0$ returns the classical game exactly. We use $\gamma = \pi/2$ as the largest nominal interpolation angle in the study; whether that endpoint is maximally entangled depends on the family and entanglement measure and is not assumed uniformly across the five families. Chappell et al. use the symbol θ for the corresponding GHZ-family angle [7]. The subscript T is the only thing left undefined here; the five entangler families it ranges over are the subject of the next subsection.

One scope limit must be stated before any equilibrium claim. Players physically hold the whole strategy space of Eq. (5), but the equilibrium tests in this paper enumerate deviations over the restricted menu $\{D, H, Q_N\}$ only. That restriction is not cosmetic: Benjamin and Hayden showed that widening the deviation space to all of SU(2) removes the equilibrium EWL originally claimed, because an ideal counter-strategy always exists [4]. Every equilibrium statement below is therefore a restricted-menu equilibrium, and we never call it a full-SU(2) result. We also work in the unitary EWL quantization rather than the alternative probabilistic scheme of Marinatto and Weber [16], which is formulated for two players only.

D. Entangler Families and Graph Symmetry

The output state, Eq. (7), is only as specific as the entangler $J_T(\gamma)$ inside it, so the five families that T ranges over must be written down before any result can be attributed to correlation structure rather than to strategy.

We compare five entangler families: GHZ, W, ring, star, and complete. They are not all the same kind of object, and the paper keeps two levels of vocabulary apart because of it. Ring, star, and complete are *graph topologies*: each is fixed by an edge set $E(G)$ on the players, and its entangler is a product of one two-qubit rotation per edge. GHZ and W carry no edge set at all. Each is a single N-body operator interpolating from the identity toward one designated entangled state, so hub, leaf, and graph-position language is undefined for them. We therefore write "entangler family" for any five-way statement and reserve "graph topology" for the pairwise three. Both global constructions are included because at three qubits GHZ and W are the two classes of genuine tripartite entanglement that stochastic local operations cannot interconvert [17], and because comparing GHZ-type against W-type correlation in N-player quantum games is an established question [7].

The GHZ family entangles the whole register with a single global rotation. We define

$$J_{\text{GHZ}}(\gamma) = \cos\frac{\gamma}{2}\, I^{\otimes N} + i\sin\frac{\gamma}{2}\, X^{\otimes N}, \qquad (9)$$

where X is the Pauli bit flip and I the single-qubit identity. Since $X^{\otimes N}$ flips every bit at once, Eq. (9) couples each classical outcome only to its bitwise complement, and in particular couples all-Dove to all-Hawk: applied to $|0^N\rangle$ at $\gamma = \pi/2$ it prepares $(|0^N\rangle + i|1^N\rangle)/\sqrt{2}$. That maximally entangled member, $(I^{\otimes N} + iX^{\otimes N})/\sqrt{2}$, is the entangling gate Benjamin and Hayden use for multiplayer quantum games [6]; Eq. (9) is the one-parameter interpolation of it that lets γ act as a control variable.

The graph topologies replace that one global rotation with one rotation per edge. For a graph G on the players we define

$$J_G(\gamma) = \prod_{(r,s)\in E(G)} \exp\Big(i\frac{\gamma}{2}X_r X_s\Big), \tag{10}$$

where X_r flips player r's qubit alone. All the operators $X_r X_s$ commute, so the product does not depend on the order in which the edges are taken and equals one exponential of their sum. The three edge sets are the cycle C_N, with $E = \{(j, j{+}1) : 1 \le j < N\} \cup \{(N, 1)\}$; the star S_N, with $E = \{(1, j) : 2 \le j \le N\}$, whose distinguished player 1 is the hub and whose remaining $N-1$ players are the leaves; and the complete graph K_N, with every pair. We index players $1, \ldots, N$ as in Sec. IV-A, while the implementation indexes qubits from zero and pins the hub at qubit 0, which is player 1 here.

Writing $m_G = |E(G)|$ for the edge count, the standard formulas are scoped to simple graphs with at least three vertices. For $N \ge 3$, the standard counts are $m_{C_N} = N$, $m_{S_N} = N - 1$, and $m_{K_N} = N(N-1)/2$. At $N = 2$, C_2 degenerates to the single edge $(1, 2)$, so $m_{C_2} = 1$; the star and complete graph are the same single edge. Thus all three graph topologies are one operator and the distinction between them is empty. At $N = 3$ the cycle C_3 is a triangle, which is K_3, so ring and complete are again the same operator and every ring-versus-complete statement in this paper is a claim about $N \ge 4$. The star is distinct from both at every $N \ge 3$ and is the only graph topology whose players are not all structurally interchangeable. Because each edge is realized by one two-qubit rotation, m_G is also the graph entangler's two-qubit gate count before routing; the counts actually executed on hardware are reported in Sec. VI-B.

The W family interpolates toward a different target state. Let $|W\rangle = N^{-1/2}\sum_{j=1}^{N} |e_j\rangle$, where e_j is the outcome in which player j alone plays Hawk, and let S_W be the reflection that exchanges $|0^N\rangle$ with $|W\rangle$:

$$\begin{aligned} S_W = I^{\otimes N} - |0^N\rangle\langle 0^N| - |W\rangle\langle W| \\ + |0^N\rangle\langle W| + |W\rangle\langle 0^N|. \end{aligned} \tag{11}$$

In words, S_W swaps the all-Dove outcome with the equal superposition of the N one-Hawk outcomes and acts as the identity on everything orthogonal to both.

Because S_W is Hermitian and squares to the identity, the interpolation that produced Eq. (9) applies to it unchanged:

$$J_W(\gamma) = \cos\frac{\gamma}{2}\, I^{\otimes N} + i \sin\frac{\gamma}{2}\, S_W, \tag{12}$$

which equals $\exp(i\gamma S_W/2)$, is unitary at every γ, and satisfies $J_W(0) = I$ like the other four families. On the all-Dove input, the live-angle state is $J_W(\gamma)|0^N\rangle = \cos(\gamma/2)|0^N\rangle + i\sin(\gamma/2)|W\rangle$; hence at $\gamma = \pi/2$ the prepared state is $(|0^N\rangle + i|W\rangle)/\sqrt{2}$, not $|W\rangle$ alone. We call $|W\rangle$ the exchanged target component. This interpolation is a construction we choose: that component is fixed by the family's name, but the generator's action beyond the $\mathrm{span}\{|0^N\rangle, |W\rangle\}$ branch is not. We adopt Eq. (11) so that GHZ and W share the same cosine–sine interpolation law on the initial-state branch. Their generators remain different operators away from that branch. Every W result reported here is therefore a statement about this construction, not about W-type entanglement in general.

Equal γ across families does not mean equal interaction strength, and a family comparison has to say so. Eq. (9) and Eq. (12) apply one rotation through γ, while Eq. (10) applies m_G of them, so at the same nominal angle the complete graph carries $N(N-1)/2$ rotations against the star's $N-1$. We define the normalized interaction-strength convention

$$J_G^{\mathrm{n}}(\gamma) = \prod_{(r,s)\in E(G)} \exp\left(i\frac{\gamma}{2m_G}X_r X_s\right), \tag{13}$$

which spreads a single unit of rotation angle γ evenly over the edges so that every family carries the same summed generator weight. Two limits of the convention belong with the definition: it equalizes total rotation angle, not entangling power or gate count, and it leaves Eqs. (9) and (12) untouched, since a single global rotation already has $m = 1$.

Every reported entangler-family comparison is computed under the unnormalized convention, with all five families evaluated at the same raw γ. We define Eq. (13) here so that the family comparison of Sec. VI-A can state exactly what it does not control for: a star-versus-complete difference at equal raw γ confounds correlation structure with coupling budget, and separating the two requires re-running the comparison under Eq. (13). That sensitivity control is not among the results reported here.

Relabeling the players acts on the graph topologies in the obvious way, and that is what makes graph position a well-defined variable rather than an artifact of naming. For a permutation σ of the players let R_σ be the unitary that moves the contents of qubit r to qubit $\sigma(r)$, so that $R_\sigma X_r R_\sigma^\dagger = X_{\sigma(r)}$ and $R_\sigma|x\rangle = |\sigma(x)\rangle$ with $\sigma(x)_{\sigma(j)} = x_j$. Conjugating Eq. (10) factor by factor carries the edge (r, s) to $(\sigma(r), \sigma(s))$ and leaves the angle alone, so

$$R_\sigma J_G(\gamma) R_\sigma^\dagger = J_{\sigma(G)}(\gamma), \tag{14}$$

where $\sigma(G)$ has edge set $\{(\sigma(r), \sigma(s)) : (r, s) \in E(G)\}$. In words, entangling and then relabeling is the same as relabeling and then entangling on the relabeled graph. The global families are the degenerate case: $X^{\otimes N}$, $|0^N\rangle$, and $|W\rangle$ are each invariant under every permutation, so $R_\sigma J_{\mathrm{GHZ}} R_\sigma^\dagger = J_{\mathrm{GHZ}}$ and $R_\sigma J_W R_\sigma^\dagger = J_W$ for all σ.

Covariance of the operator becomes covariance of expected payoffs once the players act symmetrically. We state the result directly in terms of outcome probabilities and the payoff tensor, leaving the payoff-vector notation to the next subsection.

Proposition 2 (Relabeling covariance of payoffs)**.** *Let every player use the same strategy U, let G be a graph topology, and let any noise acting on the register be a channel that commutes with R_σ. Then for every permutation σ and every player j,*

$$\sum_x p_{\sigma(G)}(x) P_{\sigma(j)}(x) = \sum_x p_G(x) P_j(x). \tag{15}$$

In particular, if σ is an automorphism of G, then any two players in the same automorphism orbit receive equal expected payoffs.

Proof. Three facts drive the argument: the all-Dove state is fixed by relabeling, $R_\sigma|0^N\rangle = |0^N\rangle$; a symmetric strategy layer commutes with relabeling, $R_\sigma \bigotimes_j U = \bigotimes_j U\, R_\sigma$; and the entangler transforms by Eq. (14). Substituting the three into the output state, Eq. (7), gives $|\psi_{\text{out}}(\sigma(G))\rangle = R_\sigma|\psi_{\text{out}}(G)\rangle$, so the outcome probabilities of Eq. (8) satisfy $p_{\sigma(G)}(\sigma(x)) = p_G(x)$. The payoff tensor, Eq. (2), depends on an outcome only through the Hawk count and the player's own bit; relabeling preserves the Hawk count, $k(\sigma(x)) = k(x)$, and carries x_j to $\sigma(x)_{\sigma(j)}$, so $P_{\sigma(j)}(\sigma(x)) = P_j(x)$. Multiplying these identities, summing over all 2^N outcomes, and changing the dummy variable from $\sigma(x)$ to x gives Eq. (15). A noise channel that commutes with R_σ leaves every step valid at the density-matrix level. The orbit claim is the case $\sigma(G) = G$. $\square$

The orbit structure of each family therefore fixes which payoff equalities must hold before any circuit is run. The ring and the complete graph are vertex-transitive, so their players form a single orbit and all N expected payoffs coincide. The star has exactly two orbits, the hub alone and the $N-1$ leaves: the leaves must be paid equally, while the hub is free to differ. GHZ and W are invariant under the full symmetric group, so their players also form one orbit. Reading structural symmetry off a graph, and keeping it distinct from the equivalences of the state that graph defines, is standard for graph-defined multiparty entanglement [18]. The proposition says nothing about how far apart two different orbits sit, which is a quantitative question; Sec. VI-A measures the hub–leaf separation, and the next subsection defines the metrics in which it is expressed.

E. Expected Payoff, Advantage, Equilibrium, and Fairness

The outcome distribution becomes economically meaningful only after we map it to player payoffs. For strategy profile $\boldsymbol{U}$ and entangler family T, we define

$$\pi_j(\boldsymbol{U};T) = \sum_x p_T(x \mid \boldsymbol{U}) P_j(x), \qquad \boldsymbol{\pi} = (\pi_1, \ldots, \pi_N). \tag{16}$$

Thus π_j averages player j's payoff tensor over all measured outcomes, and $\boldsymbol{\pi}$ records the full player-level allocation. The mean payoff introduced in Eq. (4) therefore specializes to

$$\bar{\pi}(\boldsymbol{U};T) = \frac{1}{N}\sum_{j=1}^{N} \pi_j(\boldsymbol{U};T). \tag{17}$$

The payoff vector remains the primary evidence; the mean summarizes total performance per player but cannot show who gains or loses.

Incentive compatibility requires player-level deviation checks rather than a high cooperative payoff. For a fixed entangler family T, and a fixed implementation and noise path where applicable, we suppress the T argument. With $Q_{N,-j}$ denoting the other $N-1$ players at Q_N, we define the unilateral-deviation gap

$$g_j(a) = \pi_j(Q_N^{\otimes N}) - \pi_j(a, Q_{N,-j}), \qquad a \in \{D, H\}. \tag{18}$$

The profile $Q_N^{\otimes N}$ passes the restricted-menu equilibrium test if and only if $g_j(a) \geq 0$ for every player j and both deviations $a \in \{D, H\}$. This is a test within $\{D, H, Q_N\}$, not against unrestricted SU(2); deviating to Q_N itself leaves the profile unchanged.

Two payoff comparisons answer different questions and therefore receive separate names and symbols. We define the analytic-baseline advantage and the circuit-relative payoff gap by

$$\Delta_{\text{ana}}(\boldsymbol{U};T) = \bar{\pi}(\boldsymbol{U};T) - \frac{V-C}{N}, \tag{19}$$

and

$$\Delta_{\text{circ}}(T) = \bar{\pi}(Q_N^{\otimes N};T) - \max_{\boldsymbol{a}\in\mathcal{E}_T^{D,H}} \bar{\pi}(\boldsymbol{a};T), \tag{20}$$

where $\mathcal{E}_T^{D,H}$ is the set of pure restricted $\{D,H\}^N$ circuit equilibria evaluated under the same entangler family and, where applicable, the same implementation and noise path. The first equation compares with the fixed analytic all-Hawk baseline; the second compares with a circuit-relative classical equilibrium.

Metric	Comparator
analytic-baseline advantage	Fixed analytic baseline $(V-C)/N$
circuit-relative payoff gap	Highest-mean pure restricted $\{D,H\}^N$ circuit equilibrium under the same family and path

Their values and zero crossings are not interchangeable, so we never use a bare "advantage" label when the comparator could be unclear.

State concentration is recorded separately from economic payoff. We define the all-zero target-state population by

$$P(0^N) \equiv p_T(0^N \mid \boldsymbol{U}). \tag{21}$$

This target-state population is the probability of one computational-basis outcome. For the explicit pure target $|0^N\rangle$ it equals the squared state fidelity, $F(\rho, |0^N\rangle) = \langle 0^N|\rho|0^N\rangle$.

It is not fidelity to an arbitrary ideal entangled state and does not determine the remaining populations or coherences. It is distinct from the payoff entry $P_j(x)$.

Finally, we summarize player-level dispersion only after reporting the vector. The fairness range and player floor are

$$S_\pi = \max_j \pi_j - \min_j \pi_j, \qquad F_\pi = \min_j \pi_j. \tag{22}$$

Here $S_\pi = 0$ means equal expected payoffs and F_π identifies the worst-paid player. These scalar fairness summaries do not replace the payoff vector, which remains the primary fairness evidence. With the reporting contracts fixed, the next subsection applies them to the exact GHZ cooperative benchmark.

F. GHZ Cooperative Benchmark

With the reporting contracts fixed, we now derive the cooperative reference against which unilateral deviations are measured. For the GHZ entangler, the phase structure of Q_N makes this benchmark exact at every entanglement angle, not only at maximal entanglement.

Proposition 3 (Cooperative invariance in the entanglement angle)**.** *For every γ,*

$$J^\dagger_{\mathrm{GHZ}}(\gamma)\, Q_N^{\otimes N}\, J_{\mathrm{GHZ}}(\gamma)|0^N\rangle = -|0^N\rangle, \tag{23}$$

hence $p(0^N) = 1$, $\pi_j = V/N$, and $\Delta_{\mathrm{ana}} = C/N$ independently of γ.

The minus sign is a global phase, so measurement returns the all-Dove outcome with certainty for any entanglement angle.

Proof. Using the GHZ operator of Eq. (9) and the phase gate Q_N of Eq. (6), we obtain

$$\begin{aligned} J_{\mathrm{GHZ}}(\gamma)|0^N\rangle &= \cos\Big(\frac{\gamma}{2}\Big)\,|0^N\rangle \\ &\quad + i\sin\Big(\frac{\gamma}{2}\Big)\,|1^N\rangle, \\ e^{iN\pi/N} = e^{i\pi} = -1, &\qquad e^{-iN\pi/N} = e^{-i\pi} = -1, \\ Q_N^{\otimes N}|0^N\rangle = -|0^N\rangle, &\qquad Q_N^{\otimes N}|1^N\rangle = -|1^N\rangle, \\ Q_N^{\otimes N} J_{\mathrm{GHZ}}(\gamma)|0^N\rangle &= -J_{\mathrm{GHZ}}(\gamma)|0^N\rangle, \\ J^\dagger_{\mathrm{GHZ}}(\gamma)\big[-J_{\mathrm{GHZ}}(\gamma)|0^N\rangle\big] &= -|0^N\rangle. \end{aligned}$$

The phase equalities show that both GHZ branches acquire exactly the same phase -1; the final steps show that disentangling recovers $-|0^N\rangle$. □

The payoff identities follow directly from the outcome-level contracts already fixed above. Since $p(0^N) = 1$, Eq. (2) gives $\pi_j = V/N$ for every player, and substituting the resulting mean payoff into Eq. (19) gives $\Delta_{\mathrm{ana}} = V/N-(V-C)/N = C/N$.

Corollary (live parameters). At the maximally entangled experimental benchmark, $\gamma = \pi/2$, with $V = 4$ and $C = 3$,

$$\pi_j = \frac{4}{N}, \qquad \Delta_{\mathrm{ana}} = \frac{3}{N}. \tag{24}$$

Thus every player receives $4/N$, exactly $3/N$ above the analytic all-Hawk baseline.

For the cooperative benchmark, γ is therefore a pure incentive dial: it can move the deviation incentives characterized in Sec. IV-G while leaving the cooperative payoff exactly unchanged.

G. Incentive-Compatibility Boundary

The cooperative output does not determine whether a player benefits from departing from it. We characterize the original phase choice and then introduce a distinct branch that changes its restricted incentives.

Let $Q_{N,m} = U(0, \alpha_m, \alpha_m)$, where $\alpha_m = m\pi/N$ and m is an integer. The historical strategy Q_N remains $Q_{N,1}$. Both GHZ branches acquire the common phase $(-1)^m$, so the proof in Sec. IV-F gives cooperative payoff V/N for every branch and every γ. The identity branch $m = 0$ also cooperates in outcome; that observation alone establishes no incentive compatibility.

Proposition 4 (Restricted phase-branch incentives)**.** *For $N \geq 2$, $V > 0$, $C > 0$, and $t = \sin^2\gamma \sin^2\alpha_m$, the unilateral Hawk and Dove payoffs against $Q_{N,m}$ are*

$$\begin{aligned} \pi_j(H, Q_{N,m,-j}) &= V(1-t), \\ \pi_j(D, Q_{N,m,-j}) &= (V-Ct)/N. \end{aligned} \tag{25}$$

The cooperative profile is a Nash equilibrium within $\{D, H, Q_{N,m}\}$ if and only if

$$\sin^2\gamma \sin^2\alpha_m \geq 1 - \frac{1}{N}. \tag{26}$$

The inequality requires sufficient entanglement and an appropriate cooperative phase; it is not an equilibrium condition for arbitrary SU(2) operations.

Proof. The other players contribute total phase $(N-1)\alpha_m = m\pi-\alpha_m$. For a Hawk deviation, applying $J^\dagger$ leaves support on the deviator-only Hawk outcome e_j and its complement. The probability of e_j is $|\cos^2(\gamma/2)e^{-i\alpha_m} + \sin^2(\gamma/2)e^{i\alpha_m}|^2 = 1-t$. The deviator earns V on e_j and zero on its complement. A Dove deviation leaves support on 0^N and 1^N, with probabilities $1-t$ and t, respectively, giving $(V-Ct)/N$. Comparing each payoff with V/N proves the condition, since Dove never benefits and Hawk does not benefit exactly when $t \geq 1-1/N$. □

For the original branch at $\gamma = \pi/2$, the Hawk payoff is $V\cos^2(\pi/N)$, equal to $2+2\cos(2\pi/N)$ at $V = 4$. It exceeds V/N for $N \geq 4$, reproducing the previously reported fixed-profile boundary. Define a separate alternative strategy

$$Q_N^\star = Q_{N,\lfloor N/2\rfloor}. \tag{27}$$

This selects the cooperative phase closest to $\pi/2$ from below. For even N the Hawk payoff is zero at maximal entanglement; for odd $N \geq 3$ it is $V\sin^2(\pi/(2N)) < V/N$, since

$\sin^2(\pi/(2N)) < \pi^2/(4N^2) < 1/N$. The Dove inequality is also strict. Thus $Q_N^\star$ is a strict $\{D, H, Q_N^\star\}$ equilibrium for every $N \geq 2$ in the ideal maximally entangled game. Mixtures over the same menu cannot improve on its best pure deviation. The old and alternative menus are explicitly different; no historical simulation or hardware result is relabeled.

The unrestricted boundary is explicit. Against $Q_{N,m}$, $U(\pi, 0, -\alpha_m)$ yields the sole-Hawk outcome e_j with certainty: the two amplitudes contributing to e_j have equal phase and their weights sum to $\cos^2(\gamma/2) + \sin^2(\gamma/2) = 1$. The payoff is V, a gain of $V(1 - 1/N)$, for every γ. Neither cooperative branch is therefore an equilibrium against all SU(2) deviations. This is a profile-specific witness, not a nonexistence theorem for other pure or mixed equilibria; the importance of specifying the strategy space is established in [4].

Ideal existence also differs from uniform robustness. For even N, Eq. (26) gives $\gamma_c = \arccos(1/\sqrt{N})$ on $[0, \pi/2]$, so the permitted angle window narrows as N grows, while absolute incentive margins decrease. The simulation section evaluates the alternative branch separately from the historical fixed-strategy landscape.

V. Computation and Validation Protocol

A. Exact simulation and strategy evaluation

Noiseless experiments use the dense EWL operator and exact statevector probabilities, with player j mapped to bit $j-1$ of the integer outcome. The restricted equilibrium search enumerates all 3^N profiles in the named menu and checks every unilateral change. Symmetric continuous profile searches are candidate-generation procedures. A fixed-opponent SU(2) best response is evaluated using the largest eigenvalue of a four-dimensional quadratic form, following the methodological precedent in [19]. Every player position must pass before a candidate is certified; a hub-only check is insufficient for a star.

B. Gate-level noise and compilation controls

The noisy path reconstructs each dense entangler from elementary gates, transpiles to the pinned $\{u, cx\}$ basis, and applies depolarizing channels after those gates. For a d-dimensional gate subsystem the channel is $\mathcal{D}_p(\rho) = (1 - p)\rho + pI_d/d$ when acting on that subsystem's density operator; on a larger correlated register it replaces the subsystem with I_d/d and retains the complementary reduced state. The main sweep uses equal nominal strengths on one- and two-qubit gates. The gate-level and dense operators agree up to global phase, and the $p = 0$ density-matrix output is checked against the noiseless path.

Entangler synthesis is cached by family, player count and angle. It is independent of the selected strategy profile. The reported noise ranking therefore refers to the specified decomposition and compilation settings. Matched-count and varied-

TABLE II
HARDWARE EXECUTION CONFIGURATION

Parameter	Value
Device	ibm_fez (IBM Heron r2)
Shots per circuit	4096
N=3 validation chain	$[136, 143, 142]$
Scaling chain (N=3, 4, 5)	$[59, 75, 74, 73, 79]$
Extension chain (N=3 . . . 7)	$[97, 107, 108, 109, 110, 111, 98]$
Readout mitigation	tensored confusion-matrix inversion
Zero-noise extrapolation	local cz folding, $\lambda \in \{1, 3, 5\}$
Batch sizes	11-pub scaling, 17-pub extension, 49-pub multi-topology
Software	qiskit 1.3.2 / qiskit-aer 0.14.2 / qiskit-ibm-runtime 0.36.1

compilation controls test that dependency rather than treating a circuit's gate count as an intrinsic topology invariant.

C. Uncertainty and registered decisions

Raw payoff estimates are linear functions of outcome frequencies; their multinomial covariance captures shot noise within the sampled circuit distribution. The new all-player hardware tests use simultaneous one-sided Hoeffding bounds over the registered deviation family. The exact rule and its observed outcomes appear in Sec. VI-B5. Historical readout correction and zero-noise extrapolation use their separately recorded protocols. Neither shot-noise intervals nor within-session replication establish stability across independent calibration epochs.

D. Experimental Setup and Safeguards

Every submission passes three gates: (i) an $\mathrm{Operator.equiv}$ circuit-identity assertion against the dense $J^\dagger(U^{\otimes N})J$ reference, (ii) a noiseless simulator dry-run that must reproduce the exact analytic advantage, and (iii) a full dress rehearsal of the batch, including mitigation analysis, on the device noise model reduced to the executed qubits. Job identifiers are persisted before result polling, making runs crash-safe and recoverable. The execution configuration common to all runs is summarised in Table II.

VI. Experimental Results and Analysis

A. Simulation Landscape

1) Scaling and Topology at Zero Noise: For GHZ, the fixed profile $(Q_N, \ldots, Q_N)$ pays exactly $V/N = 4/N$ per player; its circuit-evaluated restricted $\{D, H\}$ comparator is $1/N$, giving a circuit-relative payoff gap of $3/N$ (Fig. 1). The other topologies are evaluated cell by cell. Fully connected attains V/N for all recorded N = 2–6 cells; ring does so except at N=4; W attains V/N but has a larger circuit comparator and therefore a smaller gap; and star attains V/N through N=5 but deviates at N=6, with asymmetric player payoffs from N=4 onward. The restricted-menu equilibrium status is likewise cell-specific: GHZ passes the $\{D, H, Q_N\}$ deviation

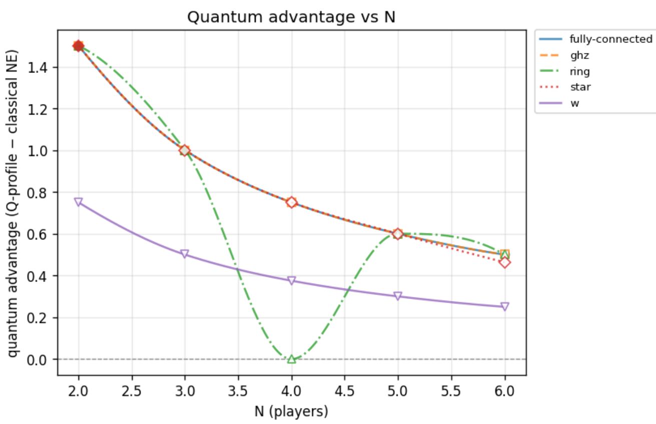

Fig. 1. Zero-noise circuit-relative payoff gap of the fixed profile $(Q_N, \ldots, Q_N)$, defined as its mean payoff minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff, at $\gamma = \pi/2$ and $V{=}4, C{=}3$. The $3/N$ gap (star markers) is attained by GHZ and fully connected at every recorded N; the ring has zero gap at $N{=}4$; the star shows a weaker $N{=}6$ value (0.463 vs. 0.5); and W tracks below throughout. Filled markers indicate that the profile passes the restricted $\{D, H, Q_N\}$ deviation check, which depends on topology.

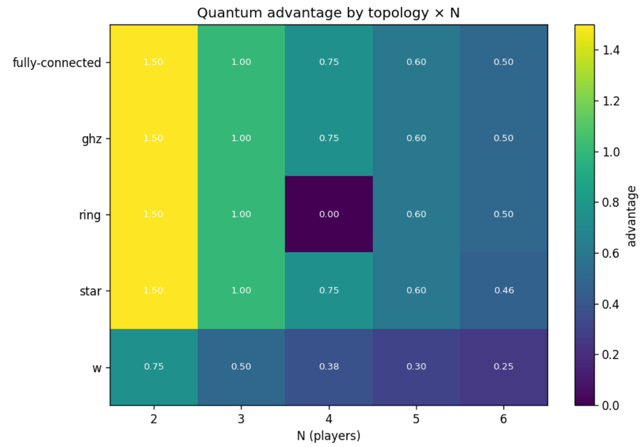

Fig. 2. Circuit-relative payoff-gap heatmap across the $(N, \text{topology})$ grid at $V{=}4, C{=}3$ (mean per-player payoff of $(Q_N, \ldots, Q_N)$ minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff). GHZ and fully connected hold the $3/N$ gap everywhere; the single dark cell is the ring at $N{=}4$ (gap exactly 0).

check at $N{=}2, 3$, and ring and fully connected also pass at $N{=}3$ in the fixed-mode artifact.

For GHZ, a separate $N = 2$–8 computation recorded in `docs/VERIFIED-FACTS.md` finds that a unilateral Hawk deviation pays $2 + 2\cos(2\pi/N)$ and that the restricted deviation result agrees with $2 + 2\cos(2\pi/N) \leq V/N$ in every computed cell. Thus at $N{=}4$ the deviation pays 2.0 against the cooperative 1.0, and at $N{=}5$ it pays $\varphi^2 \approx 2.618$ against 0.8. For $N \geq 4$ we therefore report the GHZ profile as a *fixed cooperative protocol*, not as an equilibrium prediction. The landscape also contains an $N{=}4$ ring interference zero and a star $N{=}6$ deviation from V/N; for star the circuit-relative payoff gap is asymmetric across players: at $N{=}4$ the three leaves gain 1.0 each while the hub gains nothing.

2) Entanglement-Angle Dependence: Sweeping γ separates a cooperative payoff gap from an entanglement-enabled restricted-menu equilibrium. At $\gamma = 0$, $J = I$ and the phase-only Q_N profile is measurement-equivalent to all-Dove: its positive payoff gap over all-Hawk is classical in outcome probabilities and the profile does not pass the restricted deviation check. At nonzero γ, the same sweep records when the fixed profile begins to pass that finite-menu check.

At $N{=}2$ every topology's circuit-relative payoff gap is positive from $\gamma{=}0$, and (Q, Q) begins to pass the restricted deviation check from $\gamma = 0.3\pi$ (W excepted, which never passes in range). At $N{=}3$ GHZ, fully connected, and ring hold a gap of 1.0 across the sweep and begin to pass at $\gamma = 0.3\pi$–0.4π; star and W never pass. At $N{=}4$ (Fig. 3) GHZ and star hold 0.75 at every γ, fully connected dips to 0.40 mid-sweep and recovers, W decays monotonically to 0.375, and ring reaches exactly zero at $\gamma = \pi/2$, the same $N{=}4$ interference zero. No series has a negative circuit-relative payoff gap; only

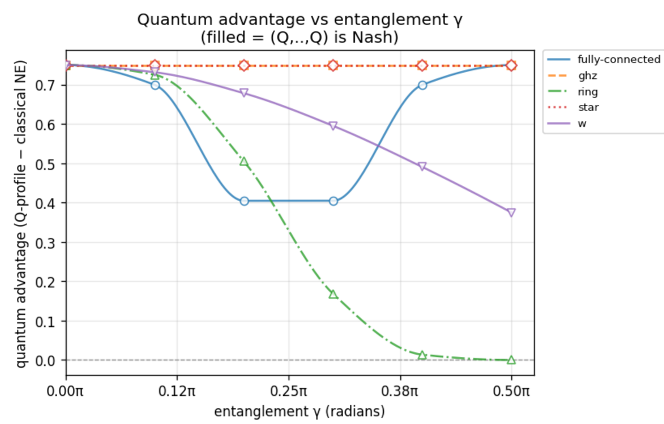

Fig. 3. Circuit-relative payoff gap versus entanglement angle γ at $N{=}4$ ($V{=}4, C{=}3$, noiseless). GHZ and star hold 0.75 flat; fully connected dips and recovers; W decays monotonically; and ring falls to zero at $\gamma = \pi/2$. Data: `results/gamma-sweep/N4`.

that ring endpoint reaches zero.

3) Player Symmetry Breaking in the Star Topology: The star is the only topology that breaks player symmetry at zero noise. For $N \geq 4$ its per-player circuit-relative payoff gaps are uneven (Fig. 4): at $N{=}4$ the three leaves have gap 1.0 each while the hub has 0.0; by $N{=}6$ the hub has 0.952 against the leaves' 0.366. The mean circuit-relative payoff gap stays close to $3/N$ until $N{=}6$, where the hub–leaf split pulls it to 0.463.

4) Noise Robustness: Two Criteria Disagree: Sweeping the depolarizing strength p from 0 to 0.05 on the gate-level noisy path, two natural "robustness" criteria give different orderings, so we state which we mean.

a) Restricted-menu equilibrium criterion: Under this criterion, the GHZ cooperative profile passes at $p = 0$ only for $N{=}2, 3$, and at $N{=}3$ it loses that status between $p = 0.04$

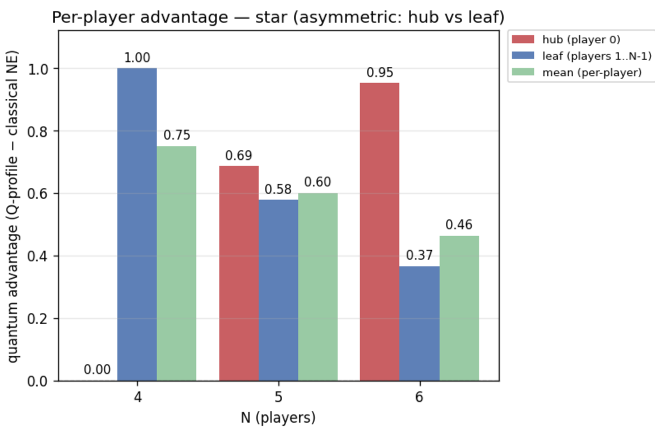

Fig. 4. Star-topology per-player circuit-relative payoff gap (hub player 0 vs. each leaf) at $\gamma = \pi/2$, $V{=}4, C{=}3$, noiseless, for $N = 4 \ldots 6$. Labels show gaps to two decimal places, including the numerically zero hub gap at $N{=}4$; green bars show the per-player mean. Data: `results/n-scaling-advantage/2026-07-19T0901Z`.

and 0.045. For $N{=}4,5$ the GHZ thresholds $p^* \approx 0.0197$ and 0.0098 are *circuit-relative payoff-gap zero crossings*, not equilibrium losses: the profile does not pass the restricted deviation check there even at $p = 0$. The fixed-mode zero-noise artifact records ring as passing at $N{=}2,3$, fully connected at $N{=}2,3$, star at $N{=}2$, and W at neither $N{=}2$ nor $N{=}3$.

b) Circuit-relative payoff-gap magnitude criterion: Under this criterion the picture flips: at $N{=}3$ GHZ retains the largest mean gap at $p = 0.05$ (0.333, 33% of its noiseless value) versus ring 0.200 and W 0.047. These are not crossings relative to the analytic $1/N$ baseline: noise can change both the fixed-profile payoff and the highest-mean restricted $\{D, H\}$ circuit equilibrium, and comparator drift can drive a zero crossing. The natural-budget ordering therefore characterizes the production circuits. Matched-count controls reduce gate-budget confounding within the XX family, whose normalized per-cx decay rates are similar in the control table at approximately 1.8–2.7; no statistical equivalence test was recorded. Absolute GHZ–W payoff-gap ordering can remain after those controls because their zero-noise gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. It keeps a positive but small mean circuit-relative gap across the whole grid, while the mean hides per-player losses: at $N{=}5$ two of the five players have negative per-player gaps from $p = 0.01$ (worst -0.077 at $p = 0.015$). All entanglers here are exact gate-level circuits; W uses the T8 construction (44 and 68 cx per J at $N{=}4,5$, versus 100/444 for the retired transpiled-unitary fallback).

5) Position-Locked Unfairness: The noise-induced per-player asymmetry is not a property of the player labels or the ideal state. A wiring-permutation test on the ring at $N{=}5$ implemented in `scripts/asymmetry_diagnostics.py` cyclically shifts the qubit-to-player wiring and the whole per-player payoff vector permutes rigidly with it (deviation $\sim 10^{-16}$); and applying the same depolarizing weight to the *ideal final state* instead of per-gate collapses the per-player spread to ~ 0. The disadvantage therefore follows the entangler's *circuit position* (gate ordering), not the ideal state or the player identity.

A distinct graph-role asymmetry was tested on hardware. A registered wiring-permutation control on the star at $N{=}4$ (Sec. VI-B4, Fig. 10C) reassigns players to different physical qubits and moves the measured per-player payoff vector by at most 0.035, inside the registered threshold, while the hub–leaf split itself (≈ 0.34 against ≈ 1.21) is unchanged. The ring simulation identifies sensitivity to circuit ordering. The star hardware control supports persistence of its hub–leaf asymmetry under the tested physical assignments; it does not identify the cause of the ring effect.

6) The Finite-Round Adaptation Pilot: An exploratory W-state pilot at $N{=}4$ tested one provisional independent round-robin best-response rule. No tested setting produced a beneficial converged fairness repair: at $p = 0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p = 0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p = 0.05$ they converged with little change. Two robustness checks were then run on this pilot, both simulation-only. Replacing round-robin with *simultaneous* best response, which removes move order entirely, reproduces all three verdicts, so the $p = 0.02$ non-convergence is a property of the noisy best-response map rather than of the update order. Extending to $N{=}5$, W $N{=}5$ reproduces the $N{=}4$ verdicts, while ring $N{=}5$ fails to converge at every noise level tested and does so in the opposite direction: it drives a perfectly fair $p{=}0$ baseline apart while leaving mean welfare unchanged, i.e. it redistributes rather than shrinks. Because those ring endpoints are snapshots of non-convergent trajectories we report the direction only and not the magnitudes. Belief-based rules such as fictitious play remain untested, and the pilot remains exploratory pending game-theory sign-off.

7) Alternative Phase and Compact Evaluation: The incentive boundary of Sec. IV-G separates a phase-choice limitation from a player-count limitation. At $N = 4, 5, 6, 7$, the alternative profile's minimum ideal Hawk/Dove gaps are respectively 0.7500, 0.4180, 0.5000, and 0.3734, whereas the original profile has a profitable Hawk deviation at each size. These are ideal results; the historical device experiments used the original phase. New tests of the alternative strategy are reported in Sec. VI-B5. The unrestricted witness remains profitable for both branches.

The ideal GHZ circuit also permits compact classical evaluation. Write $c = \cos(\gamma/2)$, $s = \sin(\gamma/2)$, $A = \bigotimes_j U_j|0\rangle$, and $B = \bigotimes_j U_j|1\rangle$. Expanding the preparation and inverse in Eq. (7) gives

$$|\psi_{\text{out}}\rangle = c^2 A + iscB - iscX^{\otimes N} A + s^2 X^{\otimes N} B. \qquad (28)$$

Thus the output is a sum of at most four product states, even for nonidentical local strategies, and its Schmidt rank across every bipartition is at most four. This is a classical representation bound for the ideal GHZ circuit, not a bound for every family or noisy implementation.

Per-player payoffs can be evaluated without enumerating outcomes using

$$\pi_j = \frac{V}{N}p(0^N) + V\,\mathbb{E}\Big[\frac{x_j}{k}\mathbf{1}_{\{k>0\}}\Big] - \frac{C}{N}p(1^N). \qquad (29)$$

For each pair of product branches, $1/k = \int_0^1 z^{k-1}\,dz$ turns the middle term into a polynomial integral of degree $N-1$. Gauss–Legendre quadrature with $\lceil N/2 \rceil$ nodes integrates it exactly up to floating point. Prefix and suffix products evaluate all players at each node in linear work, giving quadratic contraction work once the quadrature nodes are available. Nodes are cached; the timing experiment warms that cache before measurement and does not time node generation. The implementation uses no 2^N probability array or dense entangler. Tests compare arbitrary-profile payoffs with the dense oracle through $N = 7$; the resource experiment extends evaluation to $N = 128$. Timings in Fig. 5 include memory instrumentation overhead and are single-machine measurements, not asymptotic speedup evidence. Checking a fixed three-strategy candidate needs $1 + 2N$ profile evaluations; enumerating all equilibria still requires a separate search.

For a fixed profile, we also replace heuristic unilateral optimization with a global local-unitary response calculation. Quaternion quadratic-form methods already occur in two-player quantum-game analysis [19]; here we use the single-gate structure with fixed multiplayer opponents. Write $U(q) = q_0 I + i(q_1 X + q_2 Y + q_3 Z)$ with real $\|q\| = 1$. Linearity of circuit amplitudes in this single gate makes the payoff $q^T M_j q$ for a real symmetric 4×4 matrix. Four coordinate evaluations and six normalized pair sums determine M_j; its largest eigenvalue is the best-response payoff and its eigenvector supplies a witness. Independent evaluations check the reconstruction and witness. This certifies a response to fixed opponents in the specified circuit, not convergence of a search for an equilibrium. Strategy-dependent noisy compilation requires separate validation of the quadratic assumption.

8) Sensitivity to Partial Conflict Costs: The mean-welfare identity implies that analytic-baseline retention equals $1 - p(1^N)$. Uniform independent classical bits therefore retain $1 - 2^{-N}$, or 99.21875% at $N = 7$. This randomization is not an equilibrium of the underlying dominance game; it demonstrates why high retention alone does not establish coherent cooperation or useful incentive compatibility. All output patterns with between one and $N-1$ Hawks preserve the original total welfare.

We test the dependence on that payoff rule with a separate sensitivity game. Let $\ell \in [0,1]$ and define total conflict loss

$$L_\ell(k) = C\left[(1-\ell)\mathbf{1}_{\{k=N\}} + \ell\frac{k(k-1)}{N(N-1)}\right]. \qquad (30)$$

All-Dove still pays V/N; at $k > 0$ each Hawk receives $(V - L_\ell(k))/k$ and each Dove zero. The original game is $\ell = 0$, while $\ell = 1$ charges according to the fraction of Hawk pairs. Both endpoint payoffs and the $N = 2$ game are preserved. At $V > C$, Hawk remains strictly dominant since its payoff is positive against every nonempty Hawk set and is $V > V/N$ against all-Dove.

We retrospectively reweight saved raw fold-1 counts at $\ell = 0, 0.25, 0.5, 1$, without claiming new hardware execution or new equilibrium tests. For the $N = 3,\ldots,7$ extension run, retention at $\ell = 1$ is respectively 98.17%, 96.00%, 94.91%, 92.46%, and 93.23%, compared with 75% for uniform independent outcomes at every size. At $N = 7$ the corresponding raw player floor is 0.4993. Sampling intervals come from the same count distributions; they exclude calibration drift. This sensitivity model is not an empirically fitted economic model, and reweighting cooperative counts does not establish its noisy equilibria.

B. Hardware Validation on IBM Heron r2

1) Single-Point Validation at $N = 3$: We ran the validated Q_3 circuit (entangler decomposed to 6 native cz gates) with 4096 shots on ibm_fez; 3911/4096 shots landed in $|000\rangle$ (Fig. 6). The measured mean payoff is 1.3311, or 99.835% of the ideal payoff 4/3; subtracting the analytic all-Hawk baseline 1/3 gives measured advantage 0.9978, or 99.780% of the ideal advantage 1.0.

A product of the recorded readout fidelities (0.9960, 0.9961, 0.9962) and two-qubit fidelities (edge errors 0.17% and 0.23%, applied in the recorded four-plus-two CZ split) gives a heuristic ground-state probability of about 0.977, whereas the measured value is $3911/4096 = 0.955$. Under a fixed binomial interpretation the difference is about 9.5 standard deviations, so the product has the right order of magnitude but substantially underpredicts non-ground weight; it is not a calibrated statistical model of this execution.

For every outcome with $0 < k < N$ Hawks, total payoff remains V, so the mean remains V/N. Only the all-Hawk outcome lowers total payoff to $V - C$:

$$\overline{\pi} = \frac{V}{N} - \frac{C}{N}P(1\ldots1). \qquad (31)$$

In the measured counts, 176 of the 185 shots outside $|000\rangle$ land on one- and two-Hawk outcomes, while only the 9 all-Hawk shots (0.22%) lower the mean. The identity reconstructs the measured payoff 1.3311 and advantage 0.9978. The mean-payoff observable is therefore first-order insensitive to single bit-flips from the ideal outcome: errors that flip one qubit redistribute payoff between players without lowering its mean. This is the structural mechanism behind the scaling section's central observation, that the advantage decays several times slower than the all-zero target-state population, and it is why the N=3 advantage survives at 99.780% of its ideal while $P(|000\rangle)$ has already lost 4.5%.

2) Scaling at $N = 3, 4, 5$ with Error Mitigation: All three N ran on prefixes of one pinned 5-qubit chain

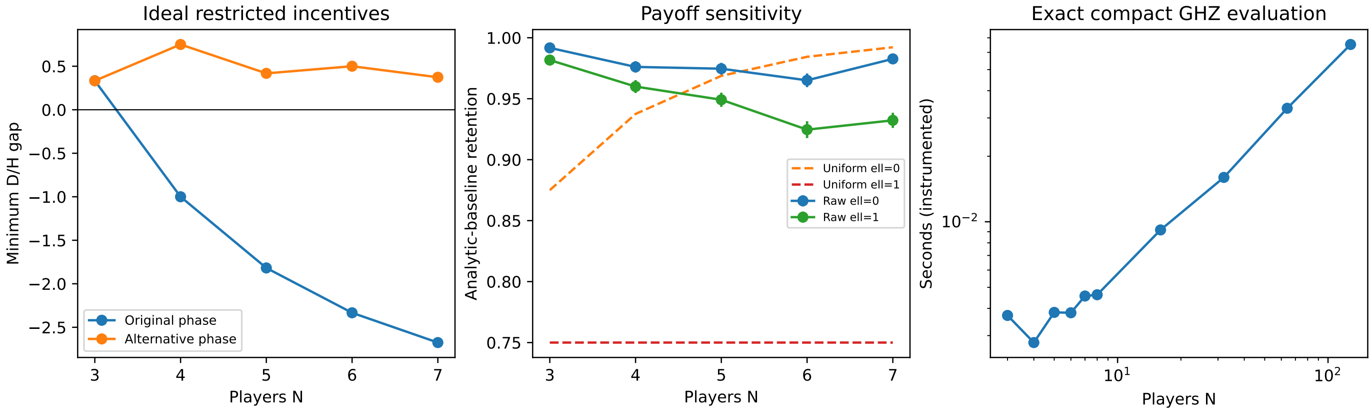

Fig. 5. Journal-extension analyses. Left: minimum ideal Hawk/Dove deviation gap for the original and alternative cooperative phase branches, each in its stated restricted menu. Middle: retrospective raw fold-1 retention under the original and partial-conflict payoff rules for the recorded $N = 3$–7 run; error bars are 1.96 multinomial standard errors and exclude calibration drift. Uniform independent-outcome comparators are diagnostic, not equilibria. Right: exact compact GHZ payoff evaluation through $N = 128$; instrumented CPU time is a classical resource measurement, not hardware player-count scaling.

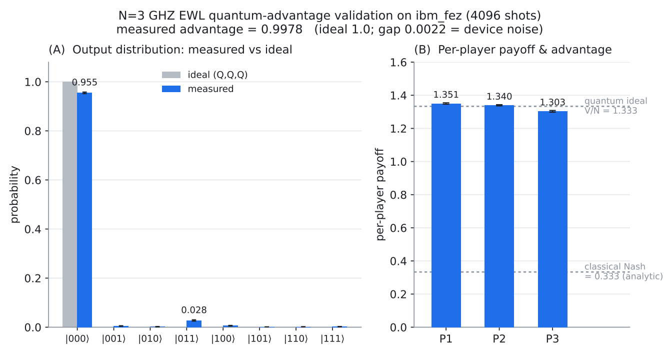

Fig. 6. Hardware validation of the N=3 GHZ EWL quantum advantage on ibm_fez (IBM Heron r2). The validated (Q, Q, Q) circuit, with entangler J decomposed to 6 native cz gates, was executed with 4096 shots. **(A)** Measured output distribution over the eight computational-basis states (blue) versus the noiseless ideal (grey); error bars are binomial shot noise. 3911/4096 shots (95.5%) fall in $|0\dots0\rangle$, the ideal (Q, Q, Q) output. **(B)** Measured per-player payoff (blue) against the analytic classical-Nash baseline (0.333, noiseless) and the quantum ideal $V/N = 1.333$; error bars propagate shot noise. The measured mean payoff 1.331 gives advantage 0.9978 vs the ideal 1.0. Physical qubits $\{136, 143, 142\}$ (calibration 2026-07-15): readout error 0.38–0.40%, cz error 136–143: 0.17%, 143–142: 0.23%.

$[59, 75, 74, 73, 79]$ (chosen by calibration error) in a single 11-pub batch: two readout-calibration circuits plus $N \in \{3, 4, 5\}\times$ cz-fold $\lambda \in \{1, 3, 5\}$. Readout errors are corrected by tensored confusion-matrix inversion (constrained least squares) [8], [9]; zero-noise extrapolation folds each cz locally (cz $\to$ cz$^{\lambda}$, unitary-preserving since cz is self-inverse) and extrapolates the readout-mitigated payoff linearly to $\lambda = 0$ [10], [11].

Calibration grouping and inclusion. The aggregate includes three executions on $[59, 75, 74, 73, 79]$, with 4096 shots per circuit, spanning two provider calibration stamps: 2026-07-16 08:30:13 +05:30 (two executions) and 2026-07-26 13:29:01 +05:30 (one execution). Executions sharing a stamp are averaged before computing the spread so that one calibration is not counted twice. A fourth execution on $[20, 21, 22, 23, 24]$ is excluded because pooling it would confound calibration drift with a change of physical qubits. Table III and Fig. 8 show the registration-source execution alone.

TABLE III
MEASURED ADVANTAGE ON IBM_FEZ FOR THE REGISTRATION-SOURCE EXECUTION (4096 SHOTS). REPEAT EXECUTIONS ARE DISCUSSED SEPARATELY; THE CROSS-CALIBRATION AGGREGATE OVER BOTH EPOCHS IS FIG. 7. IDEAL ADVANTAGE IS $3/N$; RETENTION IS ZNE/IDEAL.

N	ideal	raw	mitig.+ZNE	retention	$P(\|0\dots0\rangle)$
3	1.00	0.9919	0.9938	99.4%	0.938
4	0.75	0.7386	0.7403	98.7%	0.922
5	0.60	0.5837	0.5887	98.1%	0.868

Models and uncertainty. The ideal, device-model and depolarizing curves come from the registration anchor, not averages of refitted models. The depolarizing parameter $p_{\text{eff}} = 0.0018$ is fitted only at $N = 3$; its $N = 4, 5$ values are predictions. In Figs. 7 and 9, error bars are the sample standard deviation across the two calibration epochs. They exclude within-run multinomial and weighted-fit standard errors, and are a lower bound on total uncertainty. Figure 8 instead shows within-run payoff standard errors for the registration-source fit. Among the plotted multiplayer cases, $(Q, \dots, Q)$ passes the restricted pure-Nash check only at $N = 3$; $N = 4, 5$ measure a cooperative protocol, not an equilibrium.

The all-zero target population in Fig. 9 decays faster than the advantage: the mean-payoff observable is first-order insensitive to single bit-flips from $|0\dots0\rangle$, since a one-Hawk outcome retains mean payoff V/N. The observable mechanism is developed in Sec. VII-A.

Two model predictions frame Table III: the reduced device noise model (`NoiseModel.from_backend`) and a *fit-one-predict-two* test in the paper's own depolarizing model, with a single $p_{\text{eff}} = 0.0018$ fitted to the N=3 mitigated-fold-1 point. For the registration-source execution, the registered primary

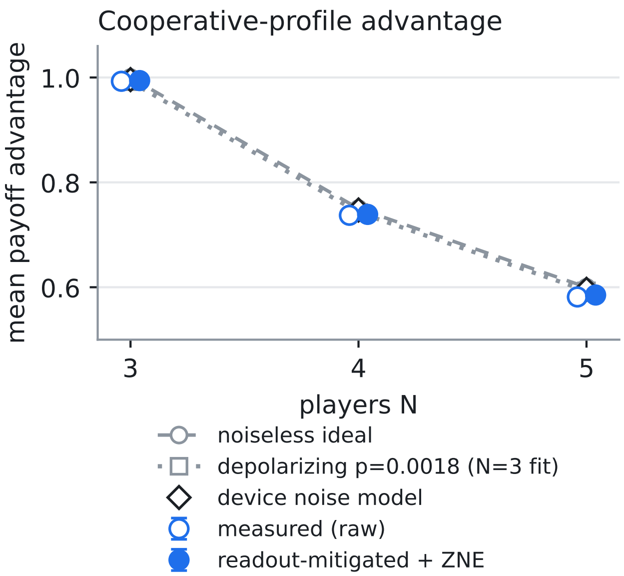

Fig. 7. Cooperative-profile payoff advantage on ibm_fez at $N = 3, 4, 5$, relative to the analytic noiseless classical-Nash baseline, comparing raw and mitigated-plus-ZNE measurements with ideal and noise-model curves. Error bars show the sample standard deviation across two calibration epochs and are a lower bound on total uncertainty; depolarizing values at $N = 4, 5$ are predictions from the $N = 3$ fit. The fixed profile passes the restricted pure-Nash check only at $N = 3$; $N = 4, 5$ measure a cooperative protocol, not an equilibrium.

mitigated-fold-1 endpoint misses at $N = 4$ by 0.0043 (2.14σ) and at $N = 5$ by 0.0101 (5.17σ). These source-run residuals are retrospective because the registration was written after that execution. For the displayed ZNE endpoint, the corresponding misses are 0.0041 (2.07σ) and 0.0060 (3.05σ). The second execution is the prospective registered repeat: its conditional $N = 4$ endpoint passes and its $N = 5$ endpoint misses by 0.0092 (4.74σ). The repeated deficit sign is observational, not a registered sign test.

The CZ-exponential baseline carries the smallest mean $|z|$ (the best score) of the five registered candidates in all four executions, including both cross-calibration repeats. It nevertheless misses $N = 5$ outside the shared 95% comparison yardstick in every run; that yardstick is for ranking, not a model-specific calibrated interval. That ranking is specific to $N \leq 5$: in the registered $N = 6, 7$ extension (Sec. VI-B3) the CZ-exponential law loses to the depolarizing model at $N{=}7$ and both fail their registered tests, so the ranking is not a physical scaling law and does not extrapolate. The registered target of three to five distinct calibrations is now met at three, and the $N = 5$ deficit reproduces with the same sign in all three repeats. The deficit magnitude is not stable across epochs, however: it ranges from 0.0092 to 0.0186, so the epoch-to-epoch variation is itself of order the effect. Model-specific predictive uncertainties remain needed, and the cross-

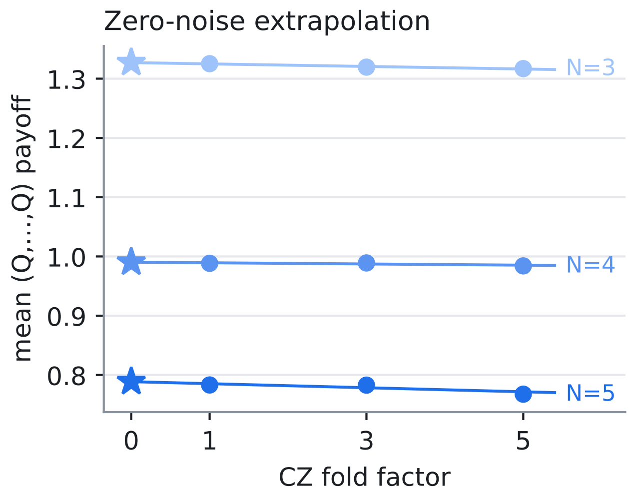

Fig. 8. Zero-noise extrapolation of the readout-mitigated mean payoff in the registration-source execution at $N = 3, 4, 5$. Points at CZ fold factors $1, 3, 5$ carry within-run payoff standard errors; weighted linear fits extrapolate to zero fold factor (stars). The $N = 4, 5$ profiles are cooperative protocols, not restricted pure-Nash equilibria.

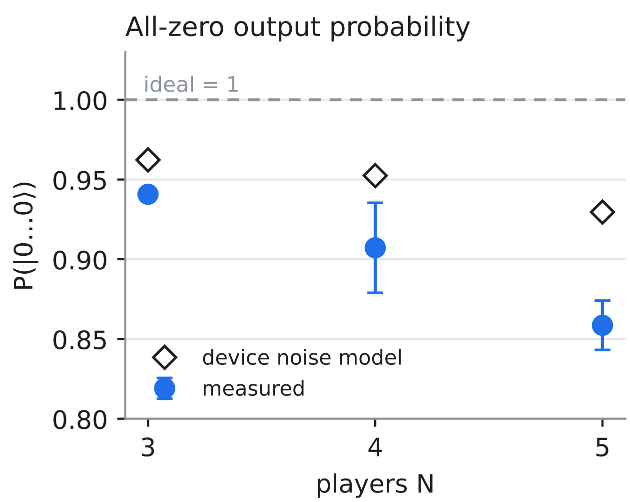

Fig. 9. All-zero output probability on ibm_fez at $N = 3, 4, 5$, compared with the device noise-model prediction and ideal target. Measured points average calibration epochs; error bars are the cross-epoch sample standard deviation and a lower bound on total uncertainty. These are fixed-profile measurements, with the restricted pure-Nash property holding only at $N = 3$.

calibration sample that is chain-matched is two epochs rather than three.

3) Extension to $N = 6, 7$: A later execution extends the curve to $N = 3, \ldots, 7$ on one pinned seven-qubit chain $[97, 107, 108, 109, 110, 111, 98]$ (17 pubs, 20 QPU-seconds). The chain was written into the registration and held at submission, so the executed qubits provably match the ones the predictions describe. Measured mitigated fold-1 advantage retains 99.2%, 97.5%, 97.3%, 96.3% and 98.1% of the ideal $3/N$ at

$N = 3, \ldots, 7$ while $P(|0\ldots0\rangle)$ falls from 0.943 to 0.784: the advantage survives to seven players at 98% of ideal with the all-zero target-state population already down to 0.78, which is the first-order-insensitivity mechanism operating across the full curve.

Two competing one-parameter models were registered before submission, both fitted on the same N=3 point and extrapolated with no $N > 3$ information: the depolarizing p_{eff} model and the CZ-exponential retention law $A_N = A_N^{\text{ideal}} r^{\,\text{cz}_N}$. *Both failed the registered* 1.96σ *test at* $N = 6, 7$, and they failed differently: the depolarizing model over-predicts at every N, while the CZ-exponential model over-predicts at $N = 4, 5, 6$ and then under-predicts at $N = 7$. A one-parameter retention law fitted at N=3 cannot reproduce that curvature.

The registered model race also split: the CZ-exponential law is closer at $N = 4, 5, 6$ but the depolarizing model is closer at $N = 7$. The CZ-exponential ranking reported below therefore does not extrapolate past N=5, and we scope it accordingly. Finally, measured advantage is strictly decreasing in N (the registered monotonicity check), but *retention* is not: N=7 keeps more of its ideal advantage than $N = 4, 5, 6$ despite carrying the most gates and the lowest target-state population, and its ZNE estimate overshoots the noiseless ideal (0.4437 against 0.4286), which no physical noise model can produce. We report this as an unexplained observation rather than a result. Distinguishing payoff geometry at odd N from an end-of-chain qubit effect or an extrapolation artefact requires a repeat *of this* $N = 3 \ldots 7$ *batch specifically*: the cross-calibration repeats aggregated in Fig. 7 re-execute the registered $N = 3, 4, 5$ batch and so cannot address it, and no such $N = 6, 7$ repeat has been run.

4) Topology, Equilibrium and Fairness on Device: The experiments above are all GHZ. To put the paper's topology, equilibrium and fairness claims on device data we ran a single 49-pub batch (4096 shots, 55 QPU-seconds) spanning five entangler topologies, unilateral strategy deviations, wiring permutations and an entanglement-angle sweep, with all predictions registered before submission. Cells were selected from *measured* routed CZ counts on the real coupling map rather than from pre-routing gate counts: on heavy-hex hardware the two differ sharply, and fully-connected N=5 (67 CZ) and W N=4, 5 (107, 219) were excluded on that basis.

The eleven cells with nonzero ideal advantage retain 97.9–99.6% of that value (Table IV, Fig. 10A), including W N=3 at 42 CZ, seven times the gate count of GHZ N=3. The mechanism therefore is not a GHZ artefact: ring N=5 and star N=5 sit at $P(|0\ldots0\rangle) \approx$ 0.10–0.11, while retaining over 99% of their analytic-baseline advantage. Their ideal outputs are not generally all-zero, so these populations alone do not measure preparation error or fidelity to their own ideal states. The topology axis does not separate these systems by how much advantage they retain; it separates them by *structure*, as the next two results show.

TABLE IV

MULTI-TOPOLOGY BATCH ON IBM_FEZ (JOB D9IA1PD0K0JC738JAQGG, 4096 SHOTS, ONE PINNED QUBIT SET). RETENTION IS MEASURED MITIGATED FOLD-1 ADVANTAGE AS A FRACTION OF *that cell's own* NOISELESS ADVANTAGE, WHICH IS NOT $3/N$ FOR EVERY TOPOLOGY.

topology	N	cz	ideal adv	measured	retention
GHZ	3	6	1.0000	0.9886	98.9%
GHZ	4	9	0.7500	0.7425	99.0%
GHZ	5	14	0.6000	0.5921	98.7%
ring	3	10	1.0000	0.9899	99.0%
ring	4	20	0.0000	0.0894	n/a
ring	5	28	0.6000	0.5960	99.3%
star	3	5	1.0000	0.9934	99.3%
star	4	9	0.7500	0.7423	99.0%
star	5	17	0.6000	0.5977	99.6%
fully-conn.	3	10	1.0000	0.9887	98.9%
fully-conn.	4	31	0.7500	0.7389	98.5%
W	3	42	1.0000	0.9792	97.9%

a) Registered Hawk-deviation tests on hardware: At GHZ N=3 the three unilateral Hawk deviation gaps measure +0.2895, +0.2726 and +0.3512 (Fig. 10B), all positive and many standard errors clear of zero. This was a registered test and it passed at every deviating position: the (Q, Q, Q) profile resists the three measured Hawk deviations. The hardware experiment does not include Dove deviations, so it does not certify the complete $\{D, H, Q_N\}$ menu under noise. It remains *not* a full-SU(2) equilibrium claim.

A registered sweep of the same gap in the entanglement angle changes sign as predicted: −0.6877 at $\gamma = 0.30\pi$ and +0.2467 at $\gamma = 0.45\pi$. The intermediate point $\gamma = 0.40\pi$ was registered in advance as marginal and excluded from the pass criterion; it returned +0.0067, statistically indistinguishable from zero. The measured Hawk-deviation incentive therefore changes sign with entanglement strength, with the transition bracketed near $\gamma = 0.4\pi$ at N=3.

b) Position-locked unfairness is a property of the topology: star N=4 is maximally unequal even noiselessly: the analytic per-player vector is $(0.25, 1.25, 1.25, 1.25)$, the hub being the loser. On device the measured vector is $(0.353, 1.202, 1.212, 1.202)$, and under two wiring permutations that reassign players to different physical qubits it moves by at most 0.035 (Fig. 10C), inside the registered threshold. The registered null therefore holds: the disadvantaged position is determined by the entanglement topology, not by which physical qubit a player was allocated. This supports the star graph-role effect under the tested assignments, separately from noise-induced ring circuit-order asymmetry.

c) A cell whose advantage is made of noise: ring N=4 is the one cell whose noiseless circuit is deterministic on $|1\ldots1\rangle$, the all-Hawk outcome, so its ideal advantage is exactly zero. It measured +0.089, the smallest of the twelve cells but clearly positive, and zero-noise extrapolation pulled it back to +0.037. All three parts of that were registered in advance. A pipeline that merely inflated whatever it touched could not produce a cell where decoherence *creates* apparent

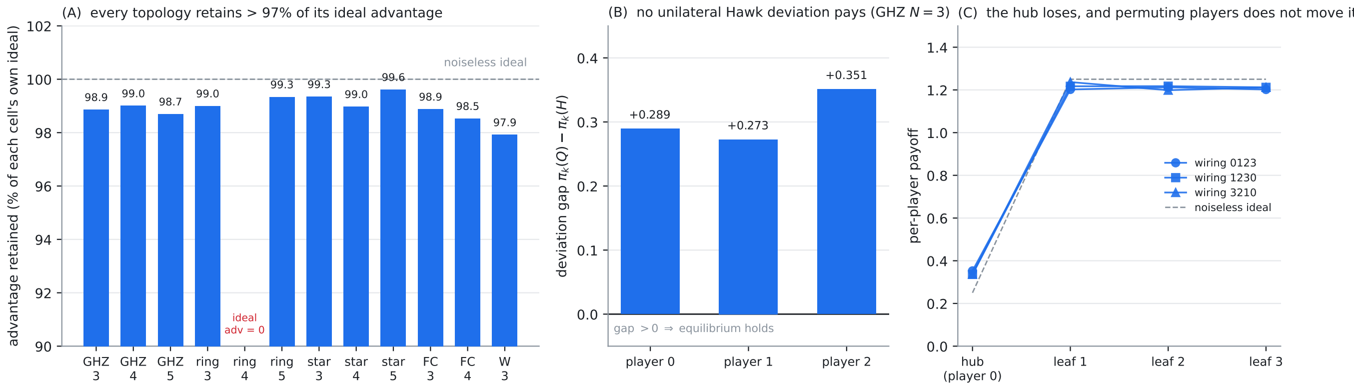

Fig. 10. Multi-topology EWL batch on ibm_fez (job `d9ia1pd0k0jc738jaqgg`), 49 pubs × 4096 shots on one pinned qubit set, all predictions registered before submission. **(A)** Measured mitigated fold-1 advantage as a percentage of each cell's *own* noiseless advantage; the eleven nonzero-baseline cells retain 97.9–99.6%. ring N=4 carries no percentage because its noiseless advantage is exactly zero: its ideal circuit is deterministic on $|1\dots1\rangle$, so the $+0.089$ it measures is created by decoherence, and zero-noise extrapolation correctly pulls it back toward zero ($+0.037$). **(B)** Unilateral Hawk deviation gaps $\pi_k(Q,Q,Q) - \pi_k(H \text{ at } k)$ at GHZ N=3: all three are positive, so no measured Hawk deviation pays; Dove deviations were not measured. **(C)** star N=4 per-player payoffs under three wirings. The hub earns ≈ 0.34 while each leaf earns ≈ 1.21, and permuting players across physical qubits does not move the pattern: the disadvantage is locked to the graph position, not to the qubit.

advantage and the extrapolation correctly removes it.

d) What did not work: The registered per-cell predictions of *absolute* advantage failed, in both the raw device-noise-model form and a CZ-linear bias-corrected form, and that comparison is additionally confounded: the provider recalibrated between registration and submission and the qubit selector moved the pinned set, so those z-scores mix model error with a change of qubits. The five results above are unaffected because each is a within-job differential: a payoff difference between two profiles measured on the same qubits in the same job, a permutation against its own twin, or one cell ranked against the other eleven. Absolute-magnitude prediction of NISQ payoffs from a device noise model remains out of reach here; the structural claims do not depend on it.

5) Registered Alternative-Phase Deviation Tests: The new experiment evaluates $Q^\star_N$ separately from the historical phase. On 7 September 2026, a committed registration fixed a seven-qubit chain, $[141, 142, 143, 144, 145, 146, 147]$, on `ibm_fez` and used its prefixes for $N = 3, \dots, 7$. An 18-circuit pilot preceded the complete 70-circuit batch. The complete batch contains a cooperative profile at each size, both Hawk and Dove deviations for every player, original-phase controls, an unrestricted counterstrategy, and two readout calibrations. Every circuit uses 4096 shots. Dense operator identity and compiled ideal output checks precede device-noise rehearsals and submission. The manifest and serialized circuits are committed before each experiment, and the execution code checks the manifest against the committed bytes.

The primary analysis uses raw implemented payoffs: calibration circuits are diagnostic and no readout correction or ZNE is applied to these tests. With $M = 50$ primary deviations and $\alpha = 0.05$, the simultaneous lower bound on each cooperative-minus-deviation gap is

$$g_j^{\mathrm{L}}(a) = \widehat{g}_j(a) - \sum_{s \in \{\mathrm{coop},\mathrm{dev}\}} V \sqrt{\frac{\log(2M/\alpha)}{2n_s}}, \tag{32}$$

where $n_s = 4096$. Hoeffding's inequality and a union bound cover the two one-sided mean errors for every gap. These bounds assume independent shots from stationary circuit distributions; they do not cover temporal drift. The registered approximate-equilibrium tolerance is $-g_j^{\mathrm{L}}(a)/(V/N) \le 0.05$ for every deviation of every player. The overall registration requires all five sizes to pass.

In the first complete batch, every measured gap is positive, but the simultaneous criterion passes only at $N = 3, 4, 5, 6$. The minimum observed gaps are 0.2360, 0.5677, 0.2157, 0.2627, and 0.0109, respectively. The corresponding maximum upper bounds on normalized deviation gain are 0.0058, -0.3240, 0.0349, -0.0285, and 0.4074. Thus the overall registered test fails: a positive point gap at $N = 7$ does not establish the requested tolerance. At $N = 4, 6$ the simultaneous lower gaps are strictly positive; at $N = 3, 5$ the supported statement is approximate equilibrium at the declared tolerance.

This all-player test locates an implementation boundary absent from mean welfare alone. At $N = 7$, target population is 0.7495 and analytic-baseline retention is 0.9600, despite unresolved incentive compatibility. The experiment certifies neither arbitrary SU(2) deviations nor an enlarged menu containing both phase families. Original-phase controls and the unrestricted witness are separately reported diagnostics. The witness's observed gain over cooperation for player zero ranges from 2.4604 to 2.7052 across the five sizes, consistent in direction with the analytic counterexample. These diagnostic point estimates do not enlarge the registered finite-menu certification scope.

TABLE V
ALTERNATIVE-PHASE ALL-PLAYER TESTS, JUDGED SEPARATELY IN EACH BATCH. $g_{\min}$ IS THE SMALLEST OBSERVED PAYOFF GAP; $u_{\max}$ IS THE LARGEST SIMULTANEOUS UPPER BOUND ON DEVIATION GAIN NORMALIZED BY V/N. THE REGISTERED TOLERANCE REQUIRES $u_{\max} \leq 0.05$.

	First batch		Repeat	
N	$g_{\min}$	$u_{\max}$	$g_{\min}$	$u_{\max}$
3	0.2360	0.0058	0.1362	0.0806
4	0.5677	-0.3240	0.5716	-0.3279
5	0.2157	0.0349	0.1861	0.0719
6	0.2627	-0.0285	0.2483	-0.0069
7	0.0109	0.4074	-0.0033	0.4322

A second complete batch reused the frozen circuits, shots and decision rule after a replication plan was committed. Each batch is judged separately; their counts are not pooled to rescue a failed criterion. The repeat supports the criterion at $N = 4, 6$, but not at $N = 3, 5, 7$ (Table V). All 50 point gaps are positive in the first batch; one is slightly negative in the repeat, at $N = 7$. This is not a statistically established profitable deviation. Both batches fail the overall five-size test, while the positive lower gaps at $N = 4, 6$ reproduce. The executions occurred minutes apart on the same chain; this is a within-session repetition, not evidence across independent calibration epochs. The pilot and two complete batches consumed $21 + 77 + 77 = 175$ seconds of provider-reported charged QPU time.

VII. DISCUSSION

A. Why the Advantage Outlives the State

Target-state population decays several times faster than the advantage (Table III): a single bit-flip from $|0 \dots 0\rangle$ yields a one-Hawk outcome whose *mean* payoff is still exactly V/N, so the mean-payoff observable is first-order insensitive to uncorrelated single flips. Readout mitigation consequently moves the mean very little; its per-player effects are mixed and remain diagnostic analyses rather than preregistered acceptance tests. The displayed ZNE estimates are higher for every series, but no recorded population analysis attributes that shift specifically to suppression of CZ-created all-Hawk weight. The structural robustness is a property of the game's payoff geometry, not of the device.

The same single-flip terms that leave the mean invariant are what create winners and losers. Under the all-Dove branch each one-Hawk outcome transfers V to a single player and zero to the rest: the mean is protected, the individual player is not, and under gate-level noise the disadvantaged position is locked to the entangler's circuit layout rather than to any coordination failure (Sec. VI-A4). In the single exploratory W-state $N{=}4$ finite-round pilot, the provisional independent round-robin best-response rule produced no beneficial converged fairness repair in any tested setting. At $p{=}0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p{=}0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p{=}0.05$ they converged with little change. The verdict survives both checks we applied to it: it is unchanged under simultaneous rather than round-robin updates, and W $N{=}5$ reproduces it. Ring $N{=}5$ does not converge at any noise level tested and fails in the opposite direction, spreading a perfectly fair baseline apart at unchanged mean welfare, reported as a direction rather than a magnitude, since those endpoints are non-convergent snapshots. No belief-based rule has been tested. The advantage we report can therefore coexist with unequal per-player outcomes; the per-player floor is a relevant diagnostic for mechanism design.

B. Scalability and Practicality

Scalability has distinct mathematical, computational, and physical meanings in this protocol. The alternative phase branch has an ideal restricted-menu equilibrium for every N, and compact GHZ evaluation reaches $N = 128$ without dense state enumeration. Neither result increases the number of simultaneous players demonstrated on hardware: the native device evidence ends at $N = 7$. The new alternative-phase batch tests every D/H deviation but does not resolve the registered tolerance at $N = 7$. Ideal per-player gains scale as C/N, incentive margins decrease with N, and the permitted entanglement-angle interval narrows. Practical scaling therefore requires explicit statistical precision and implementation budgets.

The protocol assumes a shared preparation stage and joint inverse/readout. Our centralized experiments do not demonstrate a distributed quantum market or remove the need to trust preparation, decoding, and enforcement of a permitted strategy menu. Market architectures motivate interpretations of the graph families, but are not validated deployment models. Comparisons with classical mechanisms must state which mediation and enforcement resources are available [20].

C. Limitations and Future Work

Our primary game fixes $V = 4$ and $C = 3$ in a strict-dominance regime and charges conflict cost only at the all-Hawk outcome. High mean retention is therefore compatible with random outcomes, unequal allocations, and profitable deviations. The partial-conflict sensitivity analysis tests this modeling choice retrospectively; it does not establish a realistic economic payoff model or a noisy equilibrium of the altered game.

All restricted equilibria refer to explicitly named menus. Both cooperative phase families admit a profitable unrestricted SU(2) deviation. Historical device measurements tested Hawk deviations only, and do not establish resistance to Dove deviations under hardware noise. Arbitrary quantum channels, ancillary systems, coalitions, and adaptive multi-round strategies are outside our certification scope. Symmetric strategy search also does not exclude better asymmetric profiles or other equilibria.

Entangler-family comparisons depend on the chosen generator, interpolation angle, compiled circuit, routing, and noise placement. The W results concern one specified operator; equal nominal angles do not equalize interaction budget or entangling power. The normalized interaction convention defined above remains an unexecuted sensitivity control. Noiseless star graph-role asymmetry and noise-induced ring circuit-order asymmetry are distinct effects; the star wiring experiment alone does not identify the ring's mechanism.

Depolarizing noise is a controlled model rather than a complete device description. Readout correction and zero-noise extrapolation introduce assumptions and uncertainty; extrapolated estimates can exceed physical payoff bounds. Chain-matched scaling uncertainty rests on two calibration epochs, and the original $N = 6, 7$ batch has not been repeated. Registered one-parameter models did not extrapolate reliably. The new complete alternative-phase batches are repetitions within one session, and fail their overall five-size criterion; only $N = 4, 6$ pass in both. The compact simulation results do not extend the demonstrated physical player count. The finite-round adaptation pilot remains exploratory and does not establish convergence of general learning rules.

VIII. Novelty and Technical Differentiation

Prior multiplayer EWL studies and trading experiments establish the setting in which this work belongs [2], [3], [6], [7]. Our contribution is the joint accounting of restricted incentives, payoff geometry, and implementation-dependent fairness in a specified allocation game. The phase-branch boundary explains why the historical cooperative strategy fails beyond three players and provides a different ideal restricted-menu profile, accompanied by an unrestricted counterexample.

Matched compilation controls restrict how much of a measured family ordering can be attributed to ideal correlation structure. The payoff sensitivity analysis likewise restricts how much high mean retention establishes about cooperation. The registered hardware record contains both positive differential tests and unsuccessful quantitative predictions. These components support a study of incentive and observable robustness; they do not establish computational quantum speedup or an operational quantum market.

Data and Code Availability

Source code and recorded evidence are maintained in the public Multiplayer-QHD-Game repository. A supplementary code-and-data archive accompanies the submission materials as a fixed, hashed snapshot for review. The `results/phase-validation` directories contain the committed manifests, serialized circuits, raw counts, calibration snapshots, and provider metrics for the new runs. The `results/journal-strengthening` directory records compact-evaluation and retrospective-sensitivity outputs. The claim ledger and revision log map manuscript statements to their evidence. Historical simulation dependencies and the current hardware environment are recorded separately; replay of a frozen execution uses its recorded code revision as well as its dependency versions.

IX. Conclusion

We studied how entangler family, implementation noise, and player count affect welfare and incentives in a multiplayer EWL allocation game. The original GHZ cooperative phase has a restricted-menu equilibrium only at $N = 2, 3$, while a separately defined phase branch satisfies the ideal restricted condition for every $N \geq 2$ at maximal entanglement. Both profiles admit an explicit profitable unrestricted SU(2) deviation. An all-N restricted existence statement therefore coexists with an exact strategy-access limitation.

Historical hardware experiments validate the original cooperative protocol through seven players and its resistance to measured Hawk deviations at $N = 3$. Two new complete alternative-phase batches support the registered incentive criterion at $N = 4, 6$ in both runs. The $N = 3, 5$ criterion passes only in the first batch, and $N = 7$ remains unresolved despite high mean retention. Both batches fail the overall registered five-size test. High mean retention is partly built into the all-Hawk-only conflict penalty; the uniform comparator and retrospective partial-conflict sensitivity make that dependence explicit. At $N = 7$, the partial-conflict model retains 93.23% in raw data against a 75% uniform comparator, without implying an equilibrium or a computational advantage.

Compact classical evaluation through $N = 128$ provides a scalable analysis tool, while compilation controls and calibration records delimit the physical evidence. Repetition across independent calibration epochs and explicit resource assumptions remain necessary to connect ideal incentive compatibility with a reliable implementation.

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